Shiv Nadar University - Chennai

Department of Computer Science

25CS2741 Fundamentals of Internet of Things + Lab

Date: 18/09/2026 (Week 7) **Continuous Lab Evaluation-8** II Yr/ 3rd Sem B. Tech CSE(IoT)A&B

Allotted Marks: 10 (Hardware Implementation + Code Efficiency + Viva-Voce)

Topics: CubeMX + Keil MDK + STM32 Development Board (Blue Pill/Nucleo)

STM32 HAL libraries are allowed

Supporting Documents: **1.** Board Schematics

STM32 GPIO Bring-Up and Switch-Controlled LED

Problem Statement:

Task 8A — GPIO Input/Output

Interface a push-button and LED with the STM32. Configure the corresponding GPIO pins and develop firmware so that the LED follows the switch state.

Task 8B — Event-Based Control

Modify the firmware so that each valid switch press toggles the LED state. Investigate switch bounce and implement a simple software solution to ensure one press produces one toggle.

Task 8C — Short Press vs Long Press

Modify the application to classify a button press according to how long the button remains pressed.
Use:

- **Short press:** $<$ 1 second
- **Long press:** $\geq$ 1 second

User action	System response
Short press	Toggle LED
Long press	LED enters continuous blinking mode
Next short press	Stop blinking
Long press again	Return to blinking mode

Hint : **Core HAL functions**

Purpose	HAL function	Use in this experiment
Read switch	*HAL_GPIO_ReadPin()*	Determine pressed/released state
Control LED	*HAL_GPIO_WritePin()*	Turn LED ON/OFF
Toggle LED	*HAL_GPIO_TogglePin()*	Toggle LED on short press
Millisecond time base	*HAL_GetTick()*	Measure press duration
Delay, if needed	*HAL_Delay()*	Avoid for long-press detection; can be used for simple initial testing

Important constraint

The application must not use a blocking delay to determine whether the button is a long press.

Button pressed

$\downarrow$

Record start time

$\downarrow$

Continue executing application

$\downarrow$

Button released

$\downarrow$

Record release time

$\downarrow$

$\text{duration} = \text{release} - \text{start}$

$\downarrow$

Classify event